

Joele Cossu

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EDUCATION

La Salle Cuernavaca

Bachelor's Degree in Mechatronics Engineering

Cuernavaca, MX

Sep. 2019 – Dec. 2023

CERTIFICATIONS

Scrum Fundamentals Certified (SFC) – Six Sigma Yellow Belt Professional

EXPERIENCE

Software Engineer Internship

Aug 2022 – June 2023

National Institute of Electricity and Clean Energy (INEEL)

Cuernavaca, MX

- Supported backend systems using T-SQL, PeopleSoft, and Crystal Reports while maintaining and restoring critical financial operations.
- Reconstructed business logic and reporting systems after a cybersecurity incident, ensuring continuity of institutional processes.
- Worked with relational databases (general ledger systems), strengthening data integrity through query design and table relationships.

Full Stack Software Engineer

Nov. 2024 – April 2026

ION Group

Pisa, IT

- Owned and delivered 3 full-stack applications end-to-end, engineering features and collaborating with cross-functional teams in an Agile Scrum environment using Jira.
- Maintained and refactored legacy systems through reverse engineering, improving performance, stability, and maintainability.
- Led OWASP-aligned cybersecurity improvements, mitigating vulnerabilities and strengthening system security posture.
- Redesigned UI/UX for responsiveness and usability, enhancing user engagement across platforms.
- Managed deployments, version control (Git), and packaging while contributing to an internal framework to accelerate development cycles.

INDUSTRIAL AUTOMATION PROJECTS

Automated Production Line Prototype | TIA Portal, Ladder Logic, ABB RobotStudio Feb. 2023 – Jun. 2023

- Designed and implemented a simulated automated production line, improving process flow efficiency by 25% by developing PLC logic using Siemens TIA Portal and Ladder Logic.
- Programmed and integrated an ABB industrial robot using RobotStudio, enabling synchronized pick-and-place operations with conveyor systems.
- Configured industrial communication between PLC and peripherals using Profnet, ensuring reliable signal exchange and system coordination.
- Performed system testing and commissioning (FAT simulation), identifying and resolving logic errors to achieve stable operation.

Sorting System with Vision Integration (Simulation) | PLC, Ladder Logic, Vision Systems Oct. 2022 – Jan. 2023

- Developed a sorting system prototype, increasing classification accuracy by 30% by implementing PLC-based control logic and sensor-driven decision making.
- Simulated integration of a vision system for object detection and classification, aligning inspection outputs with PLC control signals.
- Designed HMI-style monitoring interface to visualize system states, improving operator interaction and process transparency.

Industrial Conveyor Control System | TIA Portal, Ladder Logic, HMI

May 2022 – Sep. 2022

- Engineered a conveyor belt control system, reducing manual intervention by 40% by implementing structured PLC logic and automated start/stop sequences.
- Developed HMI screens for real-time monitoring and control, enhancing usability and system diagnostics.
- Applied structured troubleshooting methods to debug PLC programs, improving system reliability and fault response time.

TECHNICAL SKILLS

Industrial Automation: PLC Programming (Siemens TIA Portal, CODESYS), Ladder Diagram (LD), Structured Text (ST), Sequential Function Chart (SFC), Function Block Diagram (FBD)
Robotics & Vision: ABB RobotStudio (Programming & Simulation), Vision Systems (OpenCV)
Software & Programming: Java, Python, C/C++, SQL (PL/SQL, T-SQL), TypeScript/JavaScript
Frameworks & Tools: Angular, Spring, Node.js, Maven, JUnit, Git, GitLab, Docker, Linux, Jira, LabView
Concepts: SCADA/HMI Systems (Basics), Control Systems, Debugging & Troubleshooting, Agile Scrum, Six Sigma, ROS2 (basic)

LANGUAGES

Languages: Spanish (Native), Italian (Native), English (Fluent), French (Proficient), Mandarin (Basic - HSK1)